A fitness-independent family of simultaneous amplifiers beyond relative fitness 3/2

Alec Kriebel
Independent Researcher
me@aleckriebel.com
ORCID: 0009-0001-9320-500X

August 22, 2026

Abstract

We study fixation from a uniformly random single mutant on finite connected loopless undirected weighted graphs under both Birth–death (Bd) and death–Birth (dB) Moran updating. Let R_{sim} be the supremum of the fitness intervals $(1, R)$ on which one graph family, chosen independently of fitness, eventually amplifies fixation under both rules. Let R_{hyb} be the unique root in $(3/2, 151/100)$ of

$$R^6 - 8R^5 + 22R^4 - 30R^3 + 21R^2 - 6R + 1.$$

We construct one fitness-independent family that simultaneously amplifies every fixed $1 < r < R_{\mathrm{hyb}}$ for all sufficiently large members of the sequence. Therefore $R_{\mathrm{sim}} \geq R_{\mathrm{hyb}} = 1.5028569127905696\ldots > 3/2$.

The construction combines a large clique with dilute hub pendants and dilute, internally heavy two-vertex satellites. An exact finite-state derivation of the weak-cut limiting trace, together with uniform establishment, cleanup, and reciprocal-invasion estimates at the scale of the eventual gain, reduces the two normalized fixation corrections to explicit response functions. Optimizing their common positivity interval yields the displayed sextic. The same analysis proves that R_{hyb} is optimal among fixed positive (σ, λ) in the displayed first-order dilute pair–pendant response model; a specialization with entirely rational edge weights already crosses 3/2. Thus 3/2 is not the exact simultaneous-amplification threshold. The unrestricted value of R_{sim}, and even a finite universal upper bound, remain open.

Keywords: evolutionary graph theory; Moran process; fixation probability; simultaneous amplification; Birth–death updating; death–Birth updating.

Mathematics Subject Classification (2020): 92D15, 60J10, 05C81.

1 Introduction

Evolutionary graph theory extends the finite-population Moran process by representing sites as vertices and replacement opportunities as weighted edges (Moran, 1958; Lieberman et al., 2005; Allen et al., 2017). The order of the two elementary Moran events matters: stars and many other structures that amplify selection when birth precedes death suppress it when death precedes birth (Hindersin and Traulsen, 2015; Allen et al., 2020). Random edge weights also tend to make undirected networks less amplifying, and often suppressing, which emphasizes that useful weighting is a design problem rather than a generic effect (Bhaumik and Masuda, 2024). Death–Birth amplification is necessarily transient in fitness on every fixed noncomplete strongly connected structure (Tkadlec et al., 2020), and transient dB amplifiers have also been sought through guided spectral edge deletion (Richter, 2023). A distinct approach randomizes the update order within a single mixed Moran process (Brewster et al., 2026). Here robustness instead means that the same graph beats the well-mixed benchmark under each pure rule separately. These tensions make robustness to update order a natural design problem rather than an automatic consequence of ordinary amplification.

A population structure is a *simultaneous amplifier* at fitness r when it raises uniformly initialized fixation probability, relative to the complete graph of the same order, under both Bd and dB updating. The asymptotic problem has the quantifier order

$$\exists\{G_t\}\ \forall r \in (1, R)\ \exists t_0(r)\ \forall t \geq t_0(r).$$

The graphs and weights may depend on t, but not on r.

A complementary fixed-graph obstruction has the opposite quantifier order. No fixed finite weighted structure is a strict dB amplifier for every $r > 1$: every structure not dB-equivalent to the complete graph is suppressing above a graph-dependent threshold R_G, whereas the equivalent class ties the baseline (Kriebel, 2026a). This $\forall G\, \exists R_G$ statement supplies no graph-uniform threshold and therefore does not rule out the present $\exists\{G_t\}\, \forall r < R\, \exists t_0(r)$ order; the thresholds R_{G_t} may vary with t.

Svoboda et al. (2024) constructed simultaneous amplifiers on $1 < r < 1.2$. A preliminary unrefereed research release established $R_{\text{sim}} \geq 3/2$ with a center–triangle family, while that particular family suppresses at its endpoint (Kriebel, 2026b). The present manuscript supersedes that release in its lower-bound conclusion by crossing $3/2$. The numerical increment is small, but the conclusion is qualitative: the earlier endpoint is not the unrestricted phase boundary.

The proof has four ingredients. First, a finite Schur complement gives the rare-migration trace without an independence or branching approximation. Second, stopped birth–death comparisons control establishment and the entire path to fixation at the smaller q/C scale of the final gain. Third, the exact two-coordinate macro chain retains adverse reversals while deriving the pair gate. Finally, optimizing the two resulting response functions produces an algebraic tangency and the degree-six polynomial in the abstract. This also proves fixed-parameter optimality within the displayed first-order pair–pendant response model. No universal upper bound is inferred; the unrestricted value of R_{sim} remains open.

2 Model, baselines, and threshold

Let $G = (V, w)$ be finite, connected, loopless, undirected, and weighted:

$$w_{uv} = w_{vu} \geq 0, \qquad w_{uu} = 0, \qquad d_u := \sum_v w_{uv} > 0.$$

A state is a mutant set $S \subseteq V$. Mutants have fitness $r > 1$ and residents fitness one. Under Bd, a parent is selected proportional to fitness and then a replacement target proportional to edge weight. Under dB, a uniformly chosen target dies and then a parent is selected proportional to fitness times edge weight. Thus, with $F(S) = |V| + (r-1)|S|$,

$$P_{\mathrm{Bd}}(S, S \cup \{v\}) = \frac{r}{F(S)} \sum_{u \in S} \frac{w_{uv}}{d_u}, \qquad v \notin S, \tag{1}$$

$$P_{\mathrm{Bd}}(S, S \setminus \{v\}) = \frac{1}{F(S)} \sum_{u \notin S} \frac{w_{uv}}{d_u}, \qquad v \in S, \tag{2}$$

$$P_{\mathrm{dB}}(S, S \cup \{v\}) = \frac{1}{|V|} \frac{r \sum_{u \in S} w_{uv}}{r \sum_{u \in S} w_{uv} + \sum_{u \notin S} w_{uv}}, \qquad v \notin S, \tag{3}$$

$$P_{\mathrm{dB}}(S, S \setminus \{v\}) = \frac{1}{|V|} \frac{\sum_{u \notin S} w_{uv}}{r \sum_{u \in S} w_{uv} + \sum_{u \notin S} w_{uv}}, \qquad v \in S. \tag{4}$$

The fixation function h_U is the unique harmonic solution with boundary values zero at $\varnothing$ and one at V. We use

$$\rho_U(G, r) = \frac{1}{|V|} \sum_{v \in V} h_U(\{v\}), \qquad U \in \{\mathrm{Bd}, \mathrm{dB}\}.$$

Direct solution of the one-dimensional complete-graph count chains gives

$$\rho_{\mathrm{Bd}}(K_n, r) = \frac{1 - r^{-1}}{1 - r^{-n}}, \tag{5}$$

$$\rho_{\mathrm{dB}}(K_n, r) = \frac{n-1}{n} \frac{1 - r^{-1}}{1 - r^{-(n-1)}}. \tag{6}$$

Both converge, for fixed $r > 1$, to

$$p(r) := 1 - \frac{1}{r}.$$

Definition 1. R_{sim} is the supremum of all $R > 1$ for which one fitness-independent family $\{G_t\}$, with $|V(G_t)| \to \infty$, satisfies

$$\rho_{\mathrm{Bd}}(G_t, r) > \rho_{\mathrm{Bd}}(K_{|V(G_t)|}, r), \qquad \rho_{\mathrm{dB}}(G_t, r) > \rho_{\mathrm{dB}}(K_{|V(G_t)|}, r)$$

for every fixed $r \in (1, R)$ and every sufficiently large t.

3 The pair–pendant construction and main theorem

Let

$$P(R) = R^6 - 8R^5 + 22R^4 - 30R^3 + 21R^2 - 6R + 1$$

and let R_{hyb} be its unique root in $(3/2, 151/100)$. Define

$$\sigma_* = \frac{-R_{\mathrm{hyb}}^3 + 4R_{\mathrm{hyb}}^2 - 3R_{\mathrm{hyb}} - 1}{2(R_{\mathrm{hyb}} - 1)}, \tag{7}$$

$$\lambda_* = \frac{2(R_{\mathrm{hyb}} - 1)(1 - \sigma_*)}{1 + \sigma_*(R_{\mathrm{hyb}}^2 - 1)}. \tag{8}$$

Exact root isolation gives

$$R_{\mathrm{hyb}} = 1.5028569127905696\ldots, \quad \sigma_* = 0.1306772822870484\ldots, \quad \lambda_* = 0.7508064830318805\ldots.$$

For $t \geq 2$, put

$$q_t = t, \qquad m_t = \lfloor \lambda_* t \rfloor, \qquad C_t = t^4, \qquad n_t = C_t + m_t + 2q_t.$$

The graph contains:

1. a unit-weight clique on C_t vertices, one designated as the hub;
2. m_t unit-weight leaves adjacent only to the hub;
3. q_t disjoint two-vertex satellites, each with internal edge weight $W_t = C_t/\sigma_*$;
4. an edge of common weight $\varepsilon_t > 0$ from every satellite vertex to every clique vertex.

There are no other edges. In particular, there are no edges between distinct satellites.

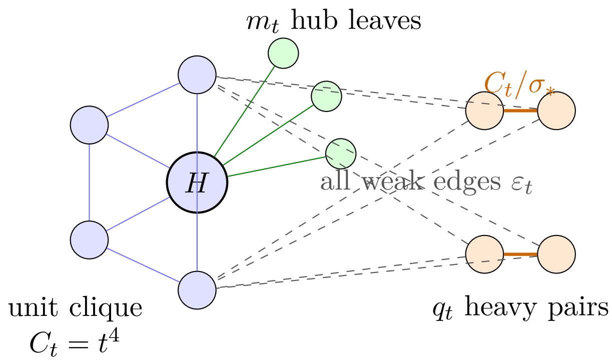

Figure 1: Schematic of the dilute hybrid. Dashed lines represent all satellite–clique edges, not only those drawn.

To define the positive coupling without reference to a subsequently chosen fitness, let

$$I_t = [1 + 1/t, R_{\mathrm{hyb}} - 1/t]$$

when this interval is nonempty. Let $\rho^0_{U,t}$ denote the finite separated trace defined in Section 4. Choose e_t to be the least positive integer such that, for $\varepsilon_t = 2^{-e_t}$,

$$\sup_{r \in I_t} \frac{n_t}{q_t} \left| \frac{\rho_U(G_t(\varepsilon_t), r)}{\rho_U(K_{n_t}, r)} - \frac{\rho^0_{U,t}(r)}{\rho_U(K_{n_t}, r)} \right| \leq \frac{1}{t} \quad (U = \mathrm{Bd}, \mathrm{dB}).$$

For the finitely many empty intervals, take $e_t = 1$.

Lemma 2 (Effective dyadic diagonal)**.** *The integer e_t exists and is computable by exact real-algebraic arithmetic. In particular, the family $\{G_t\}$ is effectively defined and every one of its weights is fixed before fitness is quantified.*

Proof. Work over the algebraic number field $\mathbb{K} = \mathbb{Q}(R_{\mathrm{hyb}})$. Both σ_* and λ_* lie in $\mathbb{K}$, and $m_t = \lfloor \lambda_* t \rfloor$ is obtained by an exact comparison of real algebraic numbers. For fixed t, e and either update rule, the transient first-step system of $G_t(2^{-e})$ has entries in $\mathbb{K}(r)$. Cramer's rule therefore puts its fixation probability in $\mathbb{K}(r)$; the same is true of the finite separated trace and the complete-graph baseline. For each $r \in I_t$, every transient block $Q(r)$ occurring in these absorbing systems is substochastic with spectral radius less than one. Hence $I - Q(r)$ is a nonsingular M-matrix and $\det(I - Q(r)) > 0$; the event denominators are positive as well.

After clearing those positive denominators, each of the two absolute-value conditions defining e_t is a pair of polynomial inequalities over $\mathbb{K}$ quantified on the algebraic interval I_t. Its truth is decidable by Sturm–subresultant arithmetic, equivalently by quantifier elimination in the real closure of $\mathbb{K}$. Proposition 5 gives uniform convergence on I_t, so some exponent is admissible. Testing $e = 1, 2, \ldots$ by the exact decision procedure therefore terminates at the least admissible e_t. $\square$

Theorem 3 (Main theorem)**.** *For every fixed $1 < r < R_{\mathrm{hyb}}$, the family above satisfies, for every sufficiently large t,*

$$\rho_{\mathrm{Bd}}(G_t, r) > \rho_{\mathrm{Bd}}(K_{n_t}, r), \qquad \rho_{\mathrm{dB}}(G_t, r) > \rho_{\mathrm{dB}}(K_{n_t}, r).$$

Consequently

$$\boxed{R_{\mathrm{sim}} \geq R_{\mathrm{hyb}} = 1.5028569127905696\ldots > 3/2.}$$

4 Exact lumping and the weak-cut limiting trace

A configuration has orbit label

$$(h, i, u, v, \ell),$$

where h is the hub type, i the number of mutant ordinary clique vertices, u the number of mixed pairs, v the number of all-mutant pairs, and ℓ the mutant-leaf count. The group

$$S_{C_t-1} \times (S_2 \wr S_{q_t}) \times S_{m_t}$$

is transitive on every fibre and commutes with both update kernels. This immediately gives the following exact reduction.

Lemma 4 (Strong orbit lumping)**.** *For every positive value of the weak-edge weight and for both update rules, the total transition probability from a labelled configuration to a target fibre depends only on (h, i, u, v, ℓ).*

Proof. Every permutation in the displayed wreath-product action preserves vertex degrees, edge weights, types, and fitnesses. It therefore commutes with each labelled parent–target event under Bd and each labelled death–parent event under dB. Summing events into a target fibre proves strong lumpability. □

We next fix finite C, m, q and send $\varepsilon \downarrow 0$. At $\varepsilon = 0$ the closed fast classes are precisely the states in which the center module and every pair are internally homogeneous. Write a macro state as

$$(h, k) \in \{0, 1\} \times \{0, \ldots, q\},$$

where h is the center type and k the number of mutant pairs. Partition the finite first-step matrix into mixed-module states F and all monomorphic-module states M, explicitly including global extinction and fixation. Their two harmonic values remain fixed at zero and one.

Proposition 5 (Finite-state weak-cut limiting trace)**.** *Let a_H^U, a_P^U be the uniformly initialized isolated fixation probabilities of the center and one pair, and let P_U^H, P_U^P be fixation probabilities of the limiting macro chain from $(1, 0)$ and $(0, 1)$. For $U \in \{\mathrm{Bd}, \mathrm{dB}\}$,*

$$\rho_U^0(C, m, q; r) = \frac{C + m}{N} a_H^U(r) P_U^H + \frac{2q}{N} a_P^U(r) P_U^P, \qquad N = C + m + 2q. \tag{9}$$

Moreover, $\rho_U(G(C, m, q, \varepsilon), r) \to \rho_U^0(C, m, q; r)$ uniformly in r on every compact subinterval of $(0, \infty)$.

Proof. Order the states as M, F. At $\varepsilon = 0$, the first-step matrix has the block form

$$P_U(0) = \begin{pmatrix} I & 0 \\ C_0 & Q_0 \end{pmatrix}, \qquad \rho(Q_0) < 1.$$

Indeed, from a mixed state each finite connected module absorbs internally almost surely, so $I - Q_0$ is invertible with nonnegative inverse and $(I - Q_0)^{-1} C_0$ is the stochastic matrix of its first monomorphic return. Expanding the positive-cut chain gives

$$P_{MM}(\varepsilon) = I + \varepsilon A + O(\varepsilon^2), \qquad P_{MF}(\varepsilon) = \varepsilon B + O(\varepsilon^2),$$

and $P_{FM}(\varepsilon) = C_0 + O(\varepsilon)$, $P_{FF}(\varepsilon) = Q_0 + O(\varepsilon)$. Eliminating F from the finite harmonic system yields, after division by the common factor ε, the macro generator

$$\mathcal{L}_U = A + B(I - Q_0)^{-1} C_0. \tag{10}$$

The second term first makes one cross-module introduction and then resolves its entire zero-cut internal excursion. Its absorption distribution is therefore exactly the affected module's local fixation probability. On the two globally absorbing macro states we retain boundary values zero and one; on every other macro state, the leading harmonic equation is $\mathcal{L}_U h = 0$. Positive introduction rates make every interior state of this finite macro ladder transient. On a compact fitness interval its interior inverse is uniformly bounded, so the boundary-value problem has a unique solution. This is the claimed introduction-rate-times-fixation trace, derived from the finite chain rather than from independent lineages.

A uniformly chosen initial mutant first resolves inside its center or pair module with probability $1 - O(\varepsilon)$. Conditioning on which module fixes locally gives (9). All matrices are finite and their entries are continuous rational functions of (ε, r) with positive event denominators. After fixing the two boundary coordinates, the scaled Schur complement converges to the invertible interior restriction of $\mathcal{L}_U$. On a compact fitness interval that limiting inverse, $(I - Q_0)^{-1}$, and the nearby absorbing inverses are uniformly bounded. The block expansion and hence the convergence are uniform there. □

5 Clique–pendant estimates at the gain scale

Let $H_{C,m}$ be a unit clique K_C with m unit hub pendants and put $p = 1 - 1/r$. Along $C = t^4$, $q = t$, and $m \asymp t$, the following estimates hold uniformly when r ranges over a compact subset of $(1, \infty)$:

Proposition 6 (Center module)**.**

$$u^U_{\mathrm{core}}(r) = p + o(q/C), \qquad U = \mathrm{Bd}, \mathrm{dB}, \tag{11}$$

$$u^{\mathrm{Bd}}_{\mathrm{leaf}}(r) = 1 - o(1), \tag{12}$$

$$u^{\mathrm{dB}}_{\mathrm{leaf}}(r) = O(C^{-1}). \tag{13}$$

Here u^U_{core} averages the C clique starting vertices. Consequently

$$a^{\mathrm{Bd}}_H(r) = \frac{Cp + m}{C + m} + o(q/C), \tag{14}$$

$$a^{\mathrm{dB}}_H(r) = \frac{Cp}{C + m} + o(q/C). \tag{15}$$

At reciprocal fitness $1/r$*,*

$$u^{\mathrm{Bd}}_{\mathrm{core}}(1/r) = o(C^{-1}), \qquad J_H(1/r) = o(C^{-1}),$$

where J_H *is the dB portal functional defined below.*

We prove the proposition through five stopped-process lemmas. This detail is needed because an $o(1)$ establishment approximation would be too weak: the final gain is only of order q/C. Fix throughout a compact interval $\mathcal{R} = [r_-, r_+] \Subset (1, \infty)$. Write a center state as (h, i, ℓ), put $c = C-1$, $d = c + m$, and suppress a common positive statewise time factor. The six Bd changing rates are

change	rate
$i \to i+1$	$r(c-i)\{h/d + i/c\}$
$i \to i-1$	$i\{(1-h)/d + (c-i)/c\}$
$0 \to 1$ at h	$r(i/c + \ell)$
$1 \to 0$ at h	$(c-i)/c + m - \ell$
$\ell \to \ell+1$	$rh(m-\ell)/d$
$\ell \to \ell-1$	$(1-h)\ell/d.$

After multiplication by $C+m$, the six dB rates are

$i \to i+1$	$(c-i)\dfrac{r(h+i)}{c+(r-1)(h+i)}$
$i \to i-1$	$i\dfrac{c-i+1-h}{c+(r-1)(i-1+h)}$
$0 \to 1$ at h	$\dfrac{r(i+\ell)}{d+(r-1)(i+\ell)}$
$1 \to 0$ at h	$\dfrac{d-i-\ell}{d+(r-1)(i+\ell)}$
$\ell \to \ell+1$	$h(m-\ell)$
$\ell \to \ell-1$	$(1-h)\ell.$

These tables follow by summing the labelled parent–target and death–parent events and are the only estimates used below.

Lemma 7 (Early ordinary-core establishment)**.** *Start from one ordinary clique mutant, with resident hub and no mutant pendants. Let* χ *be the first change of the hub or pendant coordinates, let* $K = \lceil A_0 \log C \rceil$*, and let* $\tau_j = \inf\{s : i_s = j\}$ *and* $\tau_{0,K} = \tau_0 \wedge \tau_K$*. Uniformly for* $r \in \mathcal{R}$ *and* $U \in \{\mathrm{Bd}, \mathrm{dB}\}$*,*

$$\mathbb{P}(\chi < \tau_{0,K}) = O(K^2/C), \tag{16}$$

$$\mathbb{P}_1(\tau_K < \tau_0) = 1 - \frac{1}{r} + O(K^2/C) + O(r^{-K}). \tag{17}$$

The constants are uniform for every fixed A_0*.*

Proof. Until χ, the exact embedded ordinary-count up/down odds are

$$R_{\mathrm{Bd}}(i) = r\,\frac{c-i}{c-i+c/(c+m)}, \tag{18}$$

$$R_{\mathrm{dB}}(i) = r\,\frac{c-i}{c-i+1}\,\frac{c+(r-1)(i-1)}{c+(r-1)i}. \tag{19}$$

Thus $\lambda_i^U := (R_U(i))^{-1} = r^{-1}\{1 + O(K/C)\}$ uniformly for $1 \le i < K$. For large C, the embedded ordinary walk has conditional upward probability at least $1/2 + \eta$ for some $\eta > 0$ depending only on $\mathcal{R}$. If N is its number of changes before $\{0, K\}$, optional stopping applied to the count minus its uniform positive drift gives $\mathbb{E}_1 N \le K/(2\eta) = O(K)$. At every visited state the hub-change hazard divided by the ordinary-change hazard is $O(K/C)$; a union bound over the stopped changes gives (16). Pendant coordinates cannot change before the initially resident hub does.

Suppressing that killing event, the product-odds formula is

$$\mathbb{P}_1(\tau_K < \tau_0) = \left(1 + \sum_{k=1}^{K-1} \prod_{j=1}^{k} \lambda_j^U\right)^{-1}.$$

For $k < K$, the product is $r^{-k} \exp\{O(kK/C)\}$; summing the geometric tail and restoring the killing event changes the answer by $O(K^2/C) + O(r^{-K})$. This proves (17). □

Lemma 8 (Supercritical completion and core confinement)**.** *There are constants $\delta > 0$, $r_0 > 1$, $a, b, \gamma > 0$, and an integer R_0, depending only on $\mathcal{R}$, with the following properties. Uniformly in the hub and pendant coordinates, a core family starting at $i = K$ reaches $i \ge (1-\delta)c$ before extinction except with probability*

$$O(r_0^{-K}) + e^{-\gamma C}.$$

If $R = c - i$ and $R(0) \le \delta c$, then for every fixed M,

$$\mathbb{P}\left(\sup_{s \le C^M} R_s \ge 2\delta c\right) \le C_M C^{M+1} e^{-\gamma C}, \tag{20}$$

where time is measured in the displayed continuous-time clock. Before the exceptional exit, the bounded strip is reached with the uniform tail

$$\mathbb{P}\big(\tau_{\{R \le R_0\}} > a \log C + x, \tau_{2\delta c} > \tau_{\{R \le R_0\}}\big) \le a e^{-bx}. \tag{21}$$

Proof. For every hub and pendant state, the ordinary-count odds remain bounded below by a constant $r_0 > 1$ from K to a fixed positive core density. Indeed, the two tables give the uniform lower bounds

$$\frac{q_{\mathrm{Bd}}^+}{q_{\mathrm{Bd}}^-} \ge \frac{r}{1 + c/\{d(c-i)\}}, \qquad \frac{q_{\mathrm{dB}}^+}{q_{\mathrm{dB}}^-} = r\,\frac{h+i}{i}\,\frac{c-i}{c-i+1-h}\,\frac{c+(r-1)(i-1+h)}{c+(r-1)(i+h)}.$$

For $K \to \infty$ and $i \le (1-\delta)c$, their infimum is strictly larger than one. This comparison remains valid although the hidden coordinates move. At successive i-change epochs, condition on the complete past. The pointwise intensity ratio implies that the conditional probability of an upward step is at least $r_0/(1+r_0)$: this follows equivalently by comparing the two compensator integrals up to the next i-change. Thus $r_0^{-i_n}$ is a supermartingale for the embedded adapted process. Stopping on $\{0, \lfloor (1-\delta)c \rfloor\}$ gives

$$\mathbb{P}_K\big(\tau_0 < \tau_{\lfloor (1-\delta)c \rfloor}\big) \le r_0^{-K}.$$

Applying the same conditional-bias comparison across any subsequent fixed-density interval gives an $e^{-\gamma C}$ failure bound.

In the upper strip $1 \le R \le 2\delta c$, the same tables give, uniformly in the hub and pendant coordinates,

$$\frac{q(R \mapsto R-1)}{q(R \mapsto R+1)} \ge r(1-2\delta)\frac{R}{R+1}\{1 - o(1)\}. \tag{22}$$

Writing r_- for the lower end of the chosen fitness compact, choose

$$\delta < \min\left\{\frac{1}{8}, \frac{r_- - 1}{8r_-}, \frac{r_-^2 - 1}{4(r_-^2 + 1)}\right\}.$$

Then one may take $\alpha = (r_- + 1)/2 > 1$ and, for all sufficiently large C, replace the right side of (22) by $\alpha R/(R+1)$. It is larger than $1 + \eta_0$ above the fixed integer $R_0 = \lceil(\alpha+1)/(\alpha-1)\rceil$; no statewise drift assertion is made below R_0. The hidden hub and pendant coordinates do not make the deficit marginal Markov, so we use a conditional-bias coupling rather than assuming that they do. At successive R-change epochs, condition on the entire past. Whenever $R > R_0$, the probability of an outward step divided by that of an inward step is at most $(1 + \eta_0)^{-1}$. Choose $0 < \theta < \log(1 + \eta_0)$. Then

$$\exp\{\theta R_{n\wedge\tau}\}, \qquad \tau = \tau_{\{R\le R_0\}} \wedge \tau_{\{R\ge 2\delta c\}},$$

is a supermartingale for the embedded adapted process. Optional stopping therefore gives, uniformly in all hidden-coordinate histories,

$$\sup_{k\le\delta c} \mathbb{P}_k(\tau_{\{R\ge 2\delta c\}} < \tau_{\{R\le R_0\}}) \le e^{-\theta\delta c} = e^{-\gamma C}. \tag{23}$$

The same estimate, started at $R_0 + 1$, controls every later excursion. The total R-change intensity in the displayed clock is at most ΛC. Its compensator through time C^M is therefore at most ΛC^{M+1}. Applying the strong Markov property at successive entries into $R_0 + 1$ and multiplying the last excursion bound by the expected number of entries proves (20); Markov's inequality for the compensator already suffices.

For completeness, the stopped generator calculation behind the return-time claim is also explicit. Substituting $i = c - R$ in the two tables gives

$$\begin{aligned}
\mathcal{G}_{\mathrm{Bd}}R &= (c-R)\left\{\frac{1-h}{d} + \frac{R}{c}\right\} - rR\left\{\frac{h}{d} + \frac{c-R}{c}\right\},\\
\mathcal{G}_{\mathrm{dB}}R &= (c-R)\frac{R+1-h}{c+(r-1)(c-R-1+h)} - R\frac{r(c-R+h)}{c+(r-1)(c-R+h)}.
\end{aligned}$$

After increasing R_0 and decreasing δ if necessary, both are at most $-\kappa R$ for $R_0 < R < 2\delta c$, uniformly in h, ℓ, and $r \in \mathcal{R}$. Thus $e^{\kappa(s\wedge\tau)}R_{s\wedge\tau}$ is a supermartingale in the displayed clock. On the event in (21) it has not yet hit either boundary at time $a\log C + x$. Hence

$$\mathbb{P}(\tau_{\{R\le R_0\}} > a\log C + x, \tau_{\{R\ge 2\delta c\}} > \tau_{\{R\le R_0\}}) \le \frac{\delta C}{R_0}e^{-\kappa(a\log C + x)}.$$

Taking $a > 2/\kappa$ proves (21). The states below R_0 form the announced boundary layer and require no statewise drift claim. □

Lemma 9 (Pendant synchronization and pendant initialization)**.** *For every fixed $B_0 > 0$ there is a fixed $D_{\mathrm{cl}} = D_{\mathrm{cl}}(B_0)$ such that, under either update rule and uniformly in $r \in \mathcal{R}$, a state with $i \ge (1-\delta)c$ reaches fixation within time $C^{D_{\mathrm{cl}}}$, except with probability*

$$O(C^{-B_0}) + O(C^{D_{\mathrm{cl}}+1}e^{-\gamma C}).$$

Moreover, fixation from one mutant pendant satisfies

$$u_{\mathrm{leaf}}^{\mathrm{Bd}}(r) = 1 - o(1), \qquad u_{\mathrm{leaf}}^{\mathrm{dB}}(r) = O(C^{-1}).$$

Proof. We first synchronize the hub and its pendants while the core is confined. Under Bd, freeze (i, R, ℓ) between hub and pendant events and put

$$H_+ = r(i/c + \ell), \quad H_- = R/c + m - \ell, \quad U_\ell = \frac{r(m-\ell)}{c+m}, \quad V_\ell = \frac{\ell}{c+m}.$$

If p_h is the probability that the next leaf event is upward, conditional on the current hub type h, renewal at the next hub flip gives

$$p_0 = \frac{H_+}{H_+ + V_\ell} p_1, \qquad p_1 = \frac{U_\ell}{H_- + U_\ell} + \frac{H_-}{H_- + U_\ell} p_0.$$

To make the interruption comparison formal, replace these rates by

$$\underline{H}_+ = r_-(1 - 2\delta + \ell), \qquad \overline{H}_- = 2\delta + m - \ell, \qquad \underline{U}_\ell = \frac{r_-(m-\ell)}{c+m},$$

and let $(\underline{p}_0, \underline{p}_1)$ solve the same two equations with these worst rates and V_ℓ. The resulting two-state hitting function is subharmonic for the full (h, i, ℓ) generator until the next pendant-count change: core jumps leave it unchanged, while $H_+ \geq \underline{H}_+$, $H_- \leq \overline{H}_-$, and $U_\ell \geq \underline{U}_\ell$ move its generator in the favorable direction. Indeed, the two renewal equations give

$$p_0 = \frac{H_+ U_\ell}{H_+ U_\ell + V_\ell(H_- + U_\ell)}, \qquad p_1 = \frac{U_\ell(H_+ + V_\ell)}{H_+ U_\ell + V_\ell(H_- + U_\ell)};$$

direct differentiation shows that both are increasing in H_+ and U_ℓ and decreasing in H_-. This proves the asserted favorable monotonicity rather than leaving it implicit. Optional stopping therefore makes it a lower bound even with an adapted sequence of core interruptions. Its next-leaf up/down odds, from either hub phase, are bounded below by

$$\frac{\underline{H}_+ \underline{U}_\ell}{\overline{H}_- V_\ell} \frac{1}{1 + \underline{U}_\ell / \overline{H}_-} \geq \frac{r_-^2}{1 + 2\delta}\{1 - o(1)\} > 1. \tag{24}$$

for $1 \leq \ell < m$.

The calendar-time estimate can be checked without suppressing core events. During an $h = 0$ phase the activation-or-loss intensity is bounded below by a positive constant, and during an $h = 1$ phase the deactivation-or-gain intensity is at least $m - \ell \geq 1$. Moreover,

$$\mathbb{P}(\text{loss before activation} \mid h = 0) \geq \frac{V_\ell}{r_+(1+\ell) + V_\ell} \geq \frac{c_0}{C},$$

$$\mathbb{P}(\text{gain before deactivation} \mid h = 1) \geq \frac{\underline{U}_\ell}{2\delta + m - \ell + \underline{U}_\ell} \geq \frac{c_0}{C}.$$

Core jumps contribute zero to functions of (h, ℓ) and do not pause these competing clocks. Hence the number of hub phases before the next pendant change is dominated by a geometric variable of mean $O(C)$, and each phase has uniformly bounded mean duration. Thus

$$\mathbb{E}(\sigma_{j+1} - \sigma_j \mid \mathcal{F}_{\sigma_j}) \leq K_0 C, \tag{25}$$

where σ_j are successive pendant-count changes. Thus the pendant trace is dominated below by a reflected walk whose up/down odds are at least a fixed $\beta > 1$. It makes $O(m)$ expected changes before reaching m; together with (25),

$$\sup_{R \leq 2\delta c} \mathbb{E}\tau_{\{\ell = m\}} = O(Cm). \tag{26}$$

At $\ell = m$, a resident hub activates before the next pendant loss with probability $1 - O(C^{-1})$. After activation, first suppress hub loss and keep $\ell = m$. The Bd deficit generator is then exactly

$$\mathcal{G}R = -R\left\{\frac{r}{c+m} + (r-1)\left(1 - \frac{R}{c}\right)\right\}.$$

After shrinking δ once, the bracket is at least a compact-uniform $\kappa > 0$ on $R \le 3\delta c$. Starting from $R \le 2\delta c$ and stopping at $\tau = \tau_0 \wedge \tau_{\{R \ge 3\delta c\}}$, Dynkin's formula gives

$$\mathbb{E}\int_0^\tau R_s\,ds \le \frac{R(0)}{\kappa} \le \frac{2\delta c}{\kappa}.$$

The same adapted step calculation, now with 3δ in place of 2δ, also gives $\mathbb{P}(\tau_{\{R\ge 3\delta c\}} < \tau_0) \le e^{-\gamma C}$. Conditional on the suppressed path, the probability of no hub loss is $\exp\{-\int_0^\tau R_s/c\,ds\}$. Jensen's inequality and subtraction of the upper-exit event therefore yield

$$\mathbb{P}(\text{no hub loss and } \tau_0 < \tau_{\{R\ge 3\delta c\}}) \ge e^{-2\delta/\kappa} - e^{-\gamma C} =: s_1 + o(1) > 0. \tag{27}$$

The same exponential Lyapunov estimate makes this final cleanup occur within C^2 with probability $1 - o(1)$. Together with (26) and $m = o(C)$, Markov's inequality gives a fixed $s_0 > 0$ chance of full cleanup in a block of $2C^2$ time. If hub loss ends an unsuccessful final attempt, the same adapted deficit comparison, now using the nested strips $2\delta c < 3\delta c < 4\delta c$ after shrinking δ, returns the process to $R \le \delta c$ in polynomial time before the next block; escape to $4\delta c$ before that return costs $e^{-\gamma C}$. Starting the next block from $R \le \delta c$ puts it back under the $2\delta c$ confinement estimate. The earlier hit of $3\delta c$ in (27) is charged to this same exceptional event rather than recycled as a new block. We now formalize the repetition.

Here is the stopping-time regeneration behind that last sentence. Put $\mathcal{E}_C = \{R \le \delta c\}$, write τ_{fix} for the fixation time, and let S_1 be the initial time (or the first entry into $\mathcal{E}_C$). Given S_j, run the following block: put

$$\Sigma_j = \inf\{s \ge S_j : (h_s, \ell_s) = (1, m) \text{ or } R_s \ge 2\delta c\}.$$

At each visit to $\ell = m$ with resident hub, activation precedes the next pendant loss with probability $1 - O(C^{-1})$; after a loss the same pendant trace restarts. Thus (26) also gives $\mathbb{E}(\Sigma_j - S_j \mid \mathcal{F}_{S_j}) \le KCm$, with a hit of $2\delta c$ charged as exceptional. From a nonexceptional Σ_j, run the final cleanup until fixation, hub loss, or a hit of $3\delta c$; and, after an ordinary hub-loss failure below $3\delta c$, wait for the first return to $\mathcal{E}_C$, charging a hit of $4\delta c$ as exceptional. Let T_j be the time of fixation, exceptional escape, or this return. On an ordinary return set $S_{j+1} = T_j$; a hit of $3\delta c$ is charged rather than recycled. For that return, shrink δ so that $\mathcal{G}R \le -\kappa R$ on $\delta c < R < 4\delta c$, and put $\rho = \tau_{\{R\le\delta c\}} \wedge \tau_{\{R\ge 4\delta c\}}$. The same adapted-step and generator arguments give, from $R < 3\delta c$,

$$\mathbb{P}(\tau_{\{R\ge 4\delta c\}} < \tau_{\{R\le\delta c\}}) \le e^{-\gamma C}, \qquad \mathbb{P}(\rho > t) \le 3e^{-\kappa t}.$$

Equations (26), (27), and (21), together with this nested-strip estimate, give constants $s_* > 0$, $D_* = D_*(B_0)$, and $\gamma_* > 0$ such that, uniformly on $\mathcal{F}_{S_j}$,

$$\mathbb{P}(\text{fixation during block } j,\ T_j - S_j \le C^{D_*} \mid \mathcal{F}_{S_j}) \ge s_*, \tag{28}$$

$$\mathbb{P}(\text{upper-strip escape during block } j,\ T_j - S_j \le C^{D_*} \mid \mathcal{F}_{S_j}) \le KC^{D_*+1}e^{-\gamma_* C}, \tag{29}$$

$$\mathbb{P}(T_j - S_j > C^{D_*} \mid \mathcal{F}_{S_j}) \le C^{-B_0-2}. \tag{30}$$

For the last bound, the synchronization part has conditional mean $O(Cm)$ by (26); the final attempt and return part has conditional mean $O(C^2)$ by the stopped Lyapunov estimates. Markov's inequality gives (30) after taking, for example, $D_* = B_0 + 4$. With

$$N_C = \left\lceil \frac{(B_0+2)\log C}{-\log(1-s_*)} \right\rceil,$$

the strong Markov property at the S_j gives

$$\mathbb{P}(\tau_{\mathrm{fix}} > N_C C^{D_*}) \le (1-s_*)^{N_C} + N_C\{C^{-B_0-2} + KC^{D_*+1}e^{-\gamma_* C}\} = O(C^{-B_0}).$$

Moreover $N_C C^{D_*} \le C^{D_{\mathrm{cl}}}$ for a fixed $D_{\mathrm{cl}} = D_{\mathrm{cl}}(B_0)$. This proves the claimed Bd time and failure bounds without an implicit independence assumption between blocks.

Under dB, every pendant copies the hub at its next death. In the bounded R strip, first wait for $R \le R_0$ using (21). The hub activation rate is then bounded below. During a mutant-hub interval, its deactivation intensity is at most $K(R+m-\ell)/C$. Suppressing that deactivation, direct substitution in the dB table gives $\mathcal{G}R \le -\kappa R$ while $R \le 2\delta c$, and the number $m-\ell$ of resident pendants is a pure-death chain with drift $-(m-\ell)$. Choose a fixed $\beta_0 = \beta_0(B_0) > 0$ such that

$$\beta_0 - \frac{1}{4} \ge B_0 + 2, \qquad \kappa\beta_0 \ge B_0 + 2,$$

and put $T = \beta_0 \log C$. Then

$$\begin{aligned}
\mathbb{P}(\ell_T < m) &\le me^{-T},\\
\mathbb{P}(R_T > 0,\ \tau_\uparrow > T) &\le R_0 e^{-\kappa T},\\
\mathbb{P}(\text{hub deactivates before } T\wedge\tau_\uparrow) &\le \mathbb{E}\int_0^{T\wedge\tau_\uparrow} K\frac{R_s+m-\ell_s}{C}\,ds \le \frac{K}{C}\left(\frac{R_0}{\kappa}+m\right) = o(1).
\end{aligned}$$

Here $\tau_\uparrow = \tau_{\{R\ge 2\delta c\}}$; its omitted upper-strip exit has probability $e^{-\gamma C}$. Since $m = O(C^{1/4})$, the choice of β_0 makes the first two displayed probabilities $O(C^{-B_0-2})$, whereas hub survival has probability $1-o(1)$. Thus an activated attempt started in $R \le R_0$ succeeds with probability at least $1/2$ for large C. After a failed attempt, confinement and (21) return the process to $R \le R_0$ in polynomial time; unless fixation has already occurred, the hub is activated again at a rate bounded below. The strong Markov property over $O_{B_0}(\log C)$ such attempts gives failure $O(C^{-B_0})$, while their total duration is polynomial. Taking D_{cl} larger than the two block exponents and using (20) proves the first assertion.

For a Bd mutant leaf, use the standard weighted graphical construction: each vertex u has a base reproduction clock of rate one and, when it rings, targets v with probability w_{uv}/d_u; a mutant u has an independent extra clock of rate $r-1$ with the same target law. Deleting mutants can only decrease the mutant set. Fix the original mutant pendant. At every regeneration with resident hub, no ordinary mutants, and that pendant still alive, delete all other mutants and suppress births to the other pendants. This produces the adverse lower state $(h,i,\ell) = (0,0,1)$. In its resident-hub phase, activation has rate r and loss of the distinguished pendant has rate $1/d$, so loss before activation is $O(C^{-1})$. In the ensuing mutant-hub phase, an ordinary seed has rate at least $rc/d \ge a_0$, while hub loss has rate m; hence a seed precedes hub loss with probability at least a_1/m.

After a seed, stop the ordinary count on $\{0,K\}$. For every hidden hub and pendant state, the Bd up/down odds on $1 \le i < K$ are at least a fixed $r_1 > 1$. The conditional-bias argument therefore gives $\mathbb{P}_1(\tau_K < \tau_0) \ge p_0 > 0$. The embedded walk makes $O(K)$ expected changes, and its changing intensity is bounded below, so $\mathbb{E}\tau_{0,K} = O(K)$. The distinguished-pendant loss intensity is at most $1/d$, giving $O(K/C)$ loss during this stopped trial. If the count hits zero while the pendant survives, wait for resident-hub regeneration, suppressing or deleting every ordinary reseed and every additional mutant pendant until hub loss. This is again an adverse graphical coupling: the distinguished pendant cannot be lost while its sole neighbor is mutant, and any subsequent resident-phase loss is charged as trial failure. Delete the extra mutants again at regeneration. If it hits K, the preceding completion argument fixes with probability $1 - o(1)$. Thus a trial succeeds with probability at least a_2/m, whereas it loses the distinguished pendant with probability $O((1+K)/C)$. The elementary renewal inequality for these competing outcomes gives loss before success $O(m(1+K)/C) = o(1)$. Hence $u^{\mathrm{Bd}}_{\mathrm{leaf}}(r) = 1 - o(1)$. Under dB, the leaf dies at rate one after the common death-clock time change, whereas hub activation is at most r_+/C. Fixation requires that activation, so $u^{\mathrm{dB}}_{\mathrm{leaf}}(r) = O(C^{-1})$. $\square$

Lemma 10 (Reciprocal killed-Green bounds)**.** *Put $s = 1/r$ and suppress hub changes while $h = 0, \ell = 0$. Stop the resulting ordinary-count chain I on hitting 0 or $\lceil \delta c \rceil$, and define*

$$G^U_C(i,j) = \mathbb{E}_i \int_0^\tau \mathbf{1}_{\{I_z = j\}}\, dz.$$

For every fixed L there are $a_L, \kappa, \theta > 0$, uniform for $s \in [1/r_+, 1/r_-]$ and $U \in \{\mathrm{Bd}, \mathrm{dB}\}$, such that

$$\sup_{i \le L} \mathbb{P}_i(\tau_{\lceil \delta c \rceil} < \tau_0) \le a_L e^{-\kappa C}, \tag{31}$$

$$\sup_{i \le L} \sum_j j G^U_C(i,j) \le a_L, \tag{32}$$

$$\sup_{i \le L} \sum_{j > L'} j G^U_C(i,j) \le a_L e^{-\theta L'}. \tag{33}$$

Consequently the probability of a hub change before ordinary extinction is $O_L(C^{-1})$, and the ordinary count at that change has an exponentially decreasing tail after multiplication by C.

Proof. For $i \le \delta c$, the reciprocal-fitness versions of the two rate tables give

$$\frac{q_i^+}{q_i^-} \le s_+\{1 + O(\delta) + O(C^{-1})\} \le \vartheta < 1$$

after decreasing δ, where $s_+ = 1/r_-$. They also give $q_i^- - q_i^+ \ge \kappa_0 i$ in the displayed clock. The product-odds formula for the dominated nearest-neighbor walk proves (31). Dynkin's formula for $I_{z\wedge\tau}$ gives

$$\kappa_0 \mathbb{E}_i \int_0^\tau I_z\, dz \le i,$$

which is (32). To obtain (33), let ν be the first hit of $L'+1$. Conditional bias gives $\mathbb{P}_i(\nu < \tau_0) \le K_L \vartheta^{L'-L}$. After ν, the same Dynkin estimate bounds the expected remaining particle-time by $(L'+1)/\kappa_0$. Therefore

$$\mathbb{E}_i \int_0^\tau I_s \mathbf{1}_{\{I_s > L'\}}\, ds \le \frac{K_L(L'+1)}{\kappa_0} \vartheta^{L'-L} \le a_L e^{-\theta L'},$$

after decreasing θ.

With resident hub and no mutant pendants, the hub-change intensity in either rule is at most $K_1 I_z/C$. If χ is that first change, the two precise consequences are

$$\mathbb{P}_i(\chi < \tau) \le \frac{K_1 a_L}{C}, \tag{34}$$

$$\mathbb{P}_i(\chi < \tau,\ I_\chi > L') \le \frac{K_1}{C}\mathbb{E}_i \int_0^\tau I_s \mathbf{1}_{\{I_s > L'\}}\, ds \le \frac{K_1 a_L}{C} e^{-\theta L'}. \tag{35}$$

This proves the asserted change-event bounds. □

Lemma 11 (Reciprocal hub-excursion renewal). *Uniformly for $r \in \mathcal{R}$,*

$$u^{\mathrm{Bd}}_{\mathrm{core}}(1/r) = o(C^{-1}), \qquad J_H(1/r) = o(C^{-1}).$$

Proof. For fixed L, let X_L be the states with resident hub, no mutant pendants, and at most L mutant ordinary vertices; let Y_L be the analogous states with mutant hub. Write $x^U_{C,L}$ and $y^U_{C,L}$ for the largest fixation probabilities on these two sets. Lemma 10 first gives the coarse bound

$$x^U_{C,L} \le a_L e^{-\kappa C} + a_L/C = O_L(C^{-1}). \tag{36}$$

We now make the hub-excursion stopping decomposition explicit. For Bd, start from Y_L. The hub-loss rate is at least m, while the total ordinary-changing rate is $O_L(1)$ and the pendant-seeding rate is $O(m/C)$. If hub loss is the first such event, the return state lies in X_L exactly; charging every other first event as successful fixation gives

$$y^{\mathrm{Bd}}_{C,L} \le x^{\mathrm{Bd}}_{C,L} + K_L m^{-1} + KC^{-1} = o(1). \tag{37}$$

For dB, let ζ be the first hub loss. Until the ordinary count reaches δc, the hub-loss intensity is bounded below explicitly:

$$q_{h:1\mapsto 0} = \frac{d-i-\ell}{d+(s-1)(i+\ell)} \ge \frac{(1-\delta)c}{d} =: k_h > 0.$$

While the hub is mutant, the ordinary count is dominated by a subcritical birth–death process with immigration:

$$q_i^+ \le b(i+1), \qquad q_i^- \ge d_0 i, \qquad b < d_0,$$

after decreasing δ. A common-clock monotone coupling gives $I_s \le \bar{I}_s$ for $s \le \zeta$ until the macro boundary, where $\bar{I}$ has birth rate $b(i+1)$ and death rate $d_0 i$. To make its maximum bound explicit, choose $1 < z < d_0/b$ and put $V(i) = z^i$. Its generator satisfies

$$\overline{\mathcal{G}}V(i) = (z-1)z^i\left[b - \left(\frac{d_0}{z} - b\right)i\right] \le K_z - \alpha_z V(i)$$

for suitable compact-uniform $K_z, \alpha_z > 0$. If $\bar{\tau}_M = \inf\{s : \bar{I}_s \ge M\}$, Dynkin's formula stopped at $T \wedge \bar{\tau}_M$ gives

$$\sup_{i\le L}\mathbb{P}_i\left(\sup_{s\le T}\bar{I}_s \ge M\right) \le (z^L + K_z T)z^{-M}. \tag{38}$$

Consequently the ordinary cloud before ζ is uniformly tight. Indeed, for fixed T and $L \le L_1 < \delta c$,

$$\mathbb{P}\left(\sup_{s<\zeta} I_s > L_1\right) \le e^{-k_h T} + (z^L + K_z T)z^{-L_1} =: \eta_L(L_1, T),$$

because on the event that the cloud stays below L_1 through time T, the hub-loss intensity remains at least k_h throughout. Thus the first term is precisely the surviving-hub part of the finite-T split; no infinite-horizon maximum estimate is being used. For fixed T, let $L_1 \to \infty$ and then let $T \to \infty$; the bound vanishes in that order, uniformly in C and reciprocal fitness.

On the complementary event, suppress hub reactivation after ζ. The mutant pendants are then pure deaths, so $\mathbb{E} \int_0^\infty \ell_s \, ds \le m$; the killed-Green bound gives $\mathbb{E} \int_0^\tau I_s \, ds \le a_{L_1}$. Since the hub-reactivation intensity is at most $K(I_s + \ell_s)/C$, its probability before pendant clearance and ordinary extinction is at most $K(m + a_{L_1})/C$. Conditional bias also gives

$$\mathbb{P}_{\{I \le L_1\}}(\tau_{L_2} < \tau_0) \le K_{L_1} \vartheta^{L_2 - L_1}.$$

If neither event occurs, the process enters X_{L_2}. Restoring the suppressed reactivation only adds its displayed probability, so the strong Markov property yields

$$y^{\mathrm{dB}}_{C,L} \le \eta_L(L_1, T) + \frac{K(m + a_{L_1})}{C} + K_{L_1} \vartheta^{L_2 - L_1} + x^{\mathrm{dB}}_{C,L_2}. \tag{39}$$

For fixed L_1, L_2, T, let $C \to \infty$; then let $L_2 \to \infty$, $L_1 \to \infty$, and $T \to \infty$ in that order. Equations (37)–(39) prove $y^U_{C,L} = o(1)$ for each fixed L.

Return now to one ordinary mutant. At its first hub change, the explicit hazard-weighted estimates (31) and (35) give a constant a such that

$$x^U_{C,1} \le a e^{-\kappa C} + \frac{a}{C}\big(y^U_{C,L} + e^{-\theta L}\big). \tag{40}$$

Therefore $\limsup_{C \to \infty} C x^U_{C,1} \le a e^{-\theta L}$; letting $L \to \infty$ proves $x^U_{C,1} = o(C^{-1})$, while (37)–(39) give $y^U_{C,0} = o(1)$. Hence

$$u^{\mathrm{Bd}}_{\mathrm{core}}(1/r) \le \frac{C-1}{C} x^{\mathrm{Bd}}_{C,1} + \frac{1}{C} y^{\mathrm{Bd}}_{C,0} = o(C^{-1}),$$

$$J_H(1/r) \le x^{\mathrm{dB}}_{C,1} + \frac{y^{\mathrm{dB}}_{C,0}}{c + m} = o(C^{-1}),$$

because every ordinary clique vertex has degree c and the hub has degree $c + m$. $\square$

Proof of Proposition 6. Choose A_0 so that both r^{-K} and r_0^{-K} in Lemmas 7 and 8 are $o(q/C) = o(C^{-3/4})$, and then choose $B_0 > 3/4$ in Lemma 9. Since

$$K^2/C = o(C^{-3/4}), \qquad C^{-B_0} = o(C^{-3/4}),$$

the early-establishment probability is $p + o(q/C)$ and every established family completes fixation with failure $o(q/C)$. The hub starting vertex has normalized mass $C^{-1} = o(q/C)$. This proves (11); Lemma 9 proves (12) and (13); and Lemma 11 proves the two reciprocal estimates.

Finally, average the $C - 1$ ordinary starts, the hub start, and the m pendant starts. Their contributions give (14)–(15); the exceptional hub mass is $O(C^{-1}) = o(q/C)$. $\square$

6 Pair gates and the complete post-establishment path

For the center, let

$$I_H = \sum_{x \text{ in clique}} \frac{1}{d_x} = 1 + \frac{1}{C - 1 + m}, \qquad J_H(r) = \sum_{x \text{ in clique}} \frac{h^{\mathrm{dB}}_H(\{x\}; r)}{d_x},$$

where h_H^{dB} is the isolated-center dB fixation committor. For a pair with internal weight $W = C/\sigma$, put

$$I_P = \frac{2}{W} = \frac{2\sigma}{C}, \qquad J_P(r) = \frac{1}{W} = \frac{\sigma}{C}.$$

The second identity uses the exact dB singleton value 1/2 on K_2. The center estimates imply

$$I_H = 1 + o(1), \qquad J_H(r) = p + o(1),$$

while $J_H(1/r)$ and the reciprocal core fixation probability are $o(C^{-1})$.

Proposition 12 (Gate odds and global sweep)**.** *For every fixed $r > 1$, the separated macro-fixation probabilities satisfy*

$$P_U^H = 1 - o(q/C), \qquad P_{\mathrm{Bd}}^P \longrightarrow \frac{Z_B}{1 + Z_B}, \qquad P_{\mathrm{dB}}^P \longrightarrow \frac{Z_D}{1 + Z_D},$$

where

$$Z_B = \sigma(r^2 - 1), \qquad Z_D = \frac{2r(r-1)}{\sigma}.$$

The convergence is uniform when r ranges over a compact subinterval of $(1, \infty)$.

Proof. For one specified pair, let A, D, B, C' denote, respectively, mutant-pair invasion of a resident center, resident-center recovery of that pair, mutant-center invasion of a resident pair, and resident-pair recovery of the center. In each row below a_P means the isolated pair fixation a_P^U for that update rule. Summing the parent–target or death–parent events gives, after a common time change,

	A	D	B	C'
Bd	$CrI_P u_{\mathrm{core}}^{\mathrm{Bd}}(r)$	$2I_H a_P(1/r)$	$2rI_H a_P(r)$	$CI_P u_{\mathrm{core}}^{\mathrm{Bd}}(1/r)$
dB	$2rJ_H(r)$	$(C/r)J_P(1/r)$	$CrJ_P(r)$	$(2/r)J_H(1/r)$.

The isolated pair values are

$$a_P^{\mathrm{Bd}}(r) = \frac{r}{r+1}, \quad a_P^{\mathrm{Bd}}(1/r) = \frac{1}{r+1}, \quad a_P^{\mathrm{dB}}(r) = a_P^{\mathrm{dB}}(1/r) = \frac{1}{2}.$$

On macro states the only nonzero rates are

$$\begin{array}{ll} (0,k) \to (1,k) : kA, & (0,k) \to (0,k-1) : kD, \\ (1,k) \to (1,k+1) : (q-k)B, & (1,k) \to (0,k) : (q-k)C'. \end{array}$$

From $(1,k)$, the chance that the center reverses before the next pair conversion is $C'/(B+C')$. A union bound over at most q conversions gives

$$1 - P_U^H \leq q \frac{C'}{B + C'}.$$

Proposition 6 and the portal estimates give $qC'/B = o(q/C)$, uniformly on each fitness compact. Thus a mutant center converts all resident pairs with probability $1 - o(q/C)$.

From $(0,1)$, the first successful macro event is either center invasion at rate A or loss of the sole mutant pair at rate D. Hence

$$P_U^P = \frac{A}{A+D}\, p_{1,1},$$

where $p_{1,1}$ is the macro-fixation probability from $(1,1)$. The same sweep bound, now with at most $q-1$ remaining conversions, gives $p_{1,1} \to 1$. (Equivalently, $P_U^P = [A/(A+D)][(B+C')/B]P_U^H$.) The rate table and the isolated K_2 values give

$$Z_B = \sigma(r^2-1), \qquad Z_D = \frac{2r(r-1)}{\sigma}.$$

More precisely, $A/D \to Z_U$, which yields the stated limits. The argument uses the exact absorbing macro chain, including adverse reversals; no branching survival probability is substituted for global fixation. □

7 The simultaneous correction and its optimization

Put $\eta = q/C$. Substitution of (14)–(15) and Proposition 12 into the finite trace gives the key asymptotic statement.

Proposition 13 (Two response functions)**.** *For $C = t^4$, $q = t$, $m/q \to \lambda$, and fixed $\sigma > 0$, the separated trace satisfies*

$$\frac{\rho^0_{\mathrm{Bd},t}(r)}{\rho_{\mathrm{Bd}}(K_{n_t}, r)} = 1 + \eta\left[\frac{2(\sigma-1)}{1+\sigma(r^2-1)} + \frac{\lambda}{r-1}\right] + o_r(\eta), \tag{41}$$

$$\frac{\rho^0_{\mathrm{dB},t}(r)}{\rho_{\mathrm{dB}}(K_{n_t}, r)} = 1 + \eta\left[\frac{2\{r(2-r)-\sigma\}}{\sigma+2r(r-1)} - \lambda\right] + o_r(\eta). \tag{42}$$

The remainders are uniform on every compact subinterval of $(1,\infty)$.

Proof. The complete-graph Bd baseline differs from p exponentially in C, while the dB baseline differs from p by $O(C^{-1})$; both errors are $o(\eta)$. The center terms follow from (14)–(15). The pair terms follow by the exact simplifications

$$2\left\{\frac{[r/(r+1)]Z_B/(1+Z_B)}{p} - 1\right\} = \frac{2(\sigma-1)}{1+\sigma(r^2-1)}, \tag{43}$$

$$2\left\{\frac{[1/2]Z_D/(1+Z_D)}{p} - 1\right\} = \frac{2\{r(2-r)-\sigma\}}{\sigma+2r(r-1)}. \tag{44}$$

One leaf replaces one core vertex and contributes $1/p - 1 = 1/(r-1)$ under Bd and -1 under dB. All inputs are compact-uniform by Propositions 5, 6, and 12, so the combined remainders are as stated. □

Lemma 14 (Exact tangency)**.** *At (σ_*, λ_*), both response functions in* (41)–(42) *are strictly positive for every $1 < r < R_{\mathrm{hyb}}$ and vanish simultaneously at $r = R_{\mathrm{hyb}}$.*

Proof. Both bracketed corrections are positive precisely when

$$L(r,\sigma) < \lambda < U(r,\sigma),$$

where

$$L = \frac{2(1-\sigma)(r-1)}{1+\sigma(r^2-1)}, \qquad U = \frac{2\{r(2-r)-\sigma\}}{\sigma+2r(r-1)}.$$

After multiplication by positive denominators, $L < U$ is equivalent to

$$F_r(\sigma) < 0,$$

where

$$F_r(\sigma) = (r-1)\sigma^2 + (r^3 - 4r^2 + 3r + 1)\sigma + r(2r-3).$$

Its minimizing value and minimum are

$$\sigma(r) = \frac{-r^3 + 4r^2 - 3r - 1}{2(r-1)}, \qquad \min_\sigma F_r(\sigma) = -\frac{P(r)}{4(r-1)}.$$

Exact Sturm counting gives no root of P in $(1, 3/2)$ and exactly one in $(3/2, 151/100)$. At that root, σ_* is the double tangency and

$$L(R_{\mathrm{hyb}}, \sigma_*) = U(R_{\mathrm{hyb}}, \sigma_*) = \lambda_*.$$

For fixed σ_*, L is increasing and U decreasing on $(1, R_{\mathrm{hyb}})$ because

$$\partial_r L = \frac{2(\sigma-1)\{\sigma(r-1)^2 - 1\}}{\{1 + \sigma(r^2-1)\}^2} > 0, \qquad \partial_r U = \frac{4r(\sigma - r)}{\{2r(r-1) + \sigma\}^2} < 0.$$

Here $0 < \sigma_* < 1 < r$ and $\sigma_*(r-1)^2 < 1$ throughout the isolating interval, so the displayed factorizations determine the signs without numerical approximation. Thus

$$L(r, \sigma_*) < \lambda_* < U(r, \sigma_*) \quad (1 < r < R_{\mathrm{hyb}}),$$

which proves the claim. □

Proof of Theorem 3. Fix $r \in (1, R_{\mathrm{hyb}})$. Then $r \in I_t$ for every sufficiently large t. By Proposition 13 and Lemma 14, the two separated normalized gains, after multiplication by n_t/q_t, converge to strictly positive numbers. The defining inequality for e_t changes each scaled gain by at most $1/t$. Thus both actual connected-graph gains are strictly positive for all sufficiently large t. The sequence, including every weak-edge weight, was fixed before r was chosen, which is exactly the quantifier order in the definition of R_{sim}. □

Proposition 15 (Fixed-parameter response optimality)**.** *Among fixed positive* (σ, λ) *in the displayed first-order dilute pair–pendant response model* (41)–(42)*,* R_{hyb} *is the exact largest threshold of an interval beginning at one on which both corrections can remain positive.*

Proof. For fixed positive σ, feasibility implies $F_r(\sigma) < 0$, and hence $\min_{s \in \mathbb{R}} F_r(s) < 0$. At R_{hyb} the unconstrained minimum is zero at the admissible value $0 < \sigma_* < 1$. Exact evaluation gives

$$P(3/2) = \frac{1}{64} > 0, \qquad P(151/100) = -\frac{39866792399}{10^{12}} < 0.$$

The Sturm count in Lemma 14 shows that R_{hyb} is the only root between these endpoints. Hence $P(r) < 0$ on $(R_{\mathrm{hyb}}, 151/100]$ and $\min_s F_r(s) = -P(r)/\{4(r-1)\} > 0$ there. No positive σ—indeed no real σ—is feasible immediately beyond R_{hyb}. Since the fixed pair (σ_*, λ_*) works throughout $(1, R_{\mathrm{hyb}})$, no interval beginning at one can extend farther within this response model. □

Corollary 16 (A rational-edge family)**.** *Taking* $\sigma = 19/137$ *and* $\lambda = 20/27$ *gives, exactly at* $r = 3/2$*,*

$$B = \frac{232}{17361} > 0, \qquad D = \frac{65}{12123} > 0.$$

This choice gives a family with entirely rational edge weights and algebraic response threshold

$$R_{\mathbb{Q}} = \frac{5069 + 12\sqrt{147001}}{6439} = 1.50176815223369\ldots > 3/2.$$

Thus a family with rational internal ratios and rational positive weak edges already proves a strict interval beyond $3/2$*.*

Proof. Substitution into (41)–(42) gives the two displayed margins. For these rational parameters the dB response first vanishes at the unique root in $(1, 2)$ of

$$6439r^2 - 10138r + 703 = 0,$$

which is $R_{\mathbb{Q}}$; the Bd response is still strictly positive there. Both responses are therefore positive on $(1, R_{\mathbb{Q}})$ by the same monotonicity calculation used in Lemma 14. The weak edges remain rational as well: repeat the least-dyadic diagonal construction with $I_t = [1 + 1/t, R_{\mathbb{Q}} - 1/t]$, $\sigma = 19/137$, and $m_t = \lfloor (20/27)t \rfloor$. Every edge-weight ratio in this specialized family is therefore rational. $\square$

8 Discussion

The central contribution is the quantifier in Theorem 3. One weighted graph sequence is chosen without knowing fitness and works for each fixed r in an interval extending beyond $3/2$. A family tuned separately at each fitness would establish a weaker statement. The diagonal choice of ε_t is therefore part of the construction, not merely a technical limit: compact convergence selects one positive cut that works simultaneously on an expanding interval of fitnesses.

The mechanism uses two modules with opposite response signatures. Heavy pairs contribute positively to dB and negatively to Bd, whereas hub pendants contribute positively to Bd and negatively to dB. Their densities are vanishing, so the well-mixed core supplies the baseline and the module responses appear at the common scale q_t/C_t. The exact Schur trace is important here. It resolves every local excursion before the next weak-cut event and retains the possibility that a nearly successful global sweep reverses; replacing it by an independent-lineage approximation would not prove fixation.

Biologically, Bd and dB idealize two different ways local turnover can be resolved: reproduction can select a parent before a replacement site, or a vacancy can first determine which neighbors compete. The construction shows that robustness to this order need not require most individuals to occupy a special environment. A vanishing fraction of paired niches and hub-associated peripheral sites can bias invasion in complementary ways while the bulk remains well mixed. The result is qualitative rather than a claim of empirical optimality: it identifies a mechanism by which rare structural heterogeneities can overcome update-order antagonism in this beneficial-fitness regime.

There are two sharp limitations. Proposition 15 optimizes only fixed positive (σ, λ) in the displayed first-order dilute response model. It does not classify size-dependent or singular choices σ_t, λ_t, different pair or pendant density scales, second-order gains on a vanishing first-order boundary, nonseparated pair dynamics, other modules, or non-dilute interactions. Conversely, the paper supplies no finite universal upper bound on R_{sim}. Establishing such a bound would require a comparison that couples Bd and dB fixation on arbitrary growing weighted graphs, rather than a classification of this construction.

The quantitative advantage is also asymptotically small. Here $q_t/C_t = t^{-3} = \Theta(n_t^{-3/4})$, so the normalized fixation gain tends to zero. The pair weights grow like C_t, while the exact diagonal may select very small positive weak edges. The proof gives eventual positivity for each fixed fitness but no useful finite-size bound on $t_0(r)$, and it makes no claim of empirical or finite-population optimality.

Statements and Declarations

Data and code availability. No empirical data are used. The LaTeX source, exact symbolic and rational verifiers, labelled-chain audit, and pinned build and replay scripts accompany this revision as the deterministic Supplementary Material archive `simultaneous_amplifier_beyond_three_halves_source_and_certificates.tar.gz`. It contains an internal SHA-256 manifest and a fresh-environment bootstrap. The corresponding development sources are in the paper-specific frozen repository directory. The full-repository software snapshot at doi:10.5281/zenodo.21852072 contains the unrefereed v1 manuscript, source, and software, not this superseding revision. No new persistent identifier has yet been assigned. When this revision is posted as a preprint, the same frozen bundle will be deposited alongside it and this statement updated with the assigned identifier. The replay enumerates all 512 configurations and 108 orbit fibres of its finite audit graph and independently checks the response algebra, rational margins, Sturm isolation, and tangency. These certificates audit finite transition aggregation and exact algebra; the weak-cut and population asymptotics are analytic proofs in the manuscript, not computer-assisted consequences of the replay.

Funding. No external funding supported this work.

Competing interests. The author declares no competing interests.

Author contributions. Conceptualization, formal analysis, methodology, software, validation, visualization, and writing: A.K. The author determined the scope and claims and accepts responsibility for the proof, software, and manuscript.

Ethics approval and consent. Not applicable. This study uses no human participants, animals, or empirical biological data.

AI-assisted research disclosure. Generative-AI systems were used substantively in mathematical exploration, derivation, adversarial proof analysis, software development, and manuscript preparation. The author determined the scope and claims and accepts responsibility for the manuscript and accompanying materials. All exact certificates are supplied for independent replay, and numerical evidence is explicitly separated from proof.

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